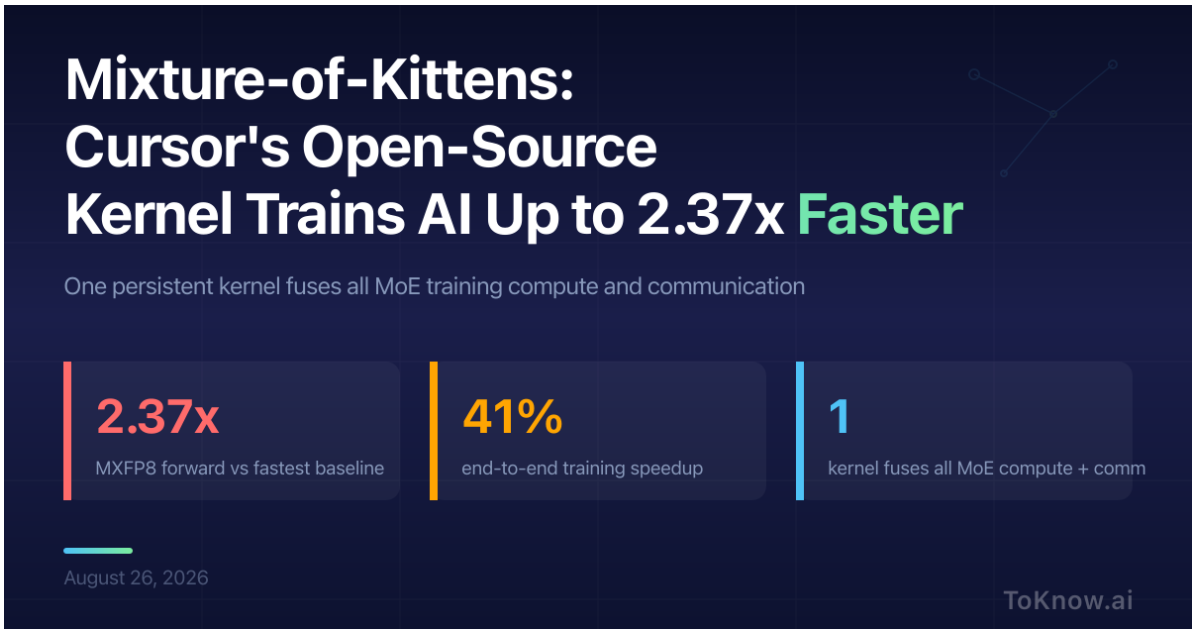
Mixture-of-Kittens: Cursor's Open-Source Kernel Trains AI Up to 2.37x Faster

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Mixture-of-Kittens (MoK) is an open-source training kernel from Cursor Research for mixture-of-experts (MoE) models, the design that spreads a model across many specialized sub-networks, called experts, and routes each input to a few of them. When experts live on different GPUs, tokens get shuttled between GPUs through many small kernels, and that overhead can eat more than half of end-to-end training time. MoK replaces that pipeline

with one persistent kernel that fuses all the computation and inter-GPU communication and removes CPU-GPU synchronization. Open-sourced under Apache-2.0 on August 4, it needs NVIDIA Blackwell NVL72 racks (CUDA 13, PyTorch 2.10) and already powers training of Cursor's Composer coding model. It is up to 2.37x faster than the fastest public baseline in 8-bit (MXFP8) precision, and 41% faster end to end on Cursor's 512-GPU production stack.

MoE is the design behind many of today's largest models: DeepSeek, GLM, Qwen, and Kimi all use the same layer pattern MoK optimizes. Because it is open source, labs without a kernel team can skip months of work. And the economics are direct: a 41% throughput gain means a three-week training run finishes in about two, on the same rented hardware, with no quality trade-off.

Cursor says AI coding agents helped its small team write the megakernel from scratch, work that a year ago needed a purpose-built framework. If agents can now do kernel engineering, the next round of training-cost wins may come as much from software as from new chips.

Read More: TurboFieldfare, Deltafin, and WASTE: three ways to run MoE models from disk

Sources:

- Mixture-of-Kittens: our open-source MoE megakernel for NVL72s (Cursor)
- Mixture-of-Kittens repository (GitHub)
- Composer 2 Technical Report (arXiv)
- Introducing Composer 2 (Cursor)

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